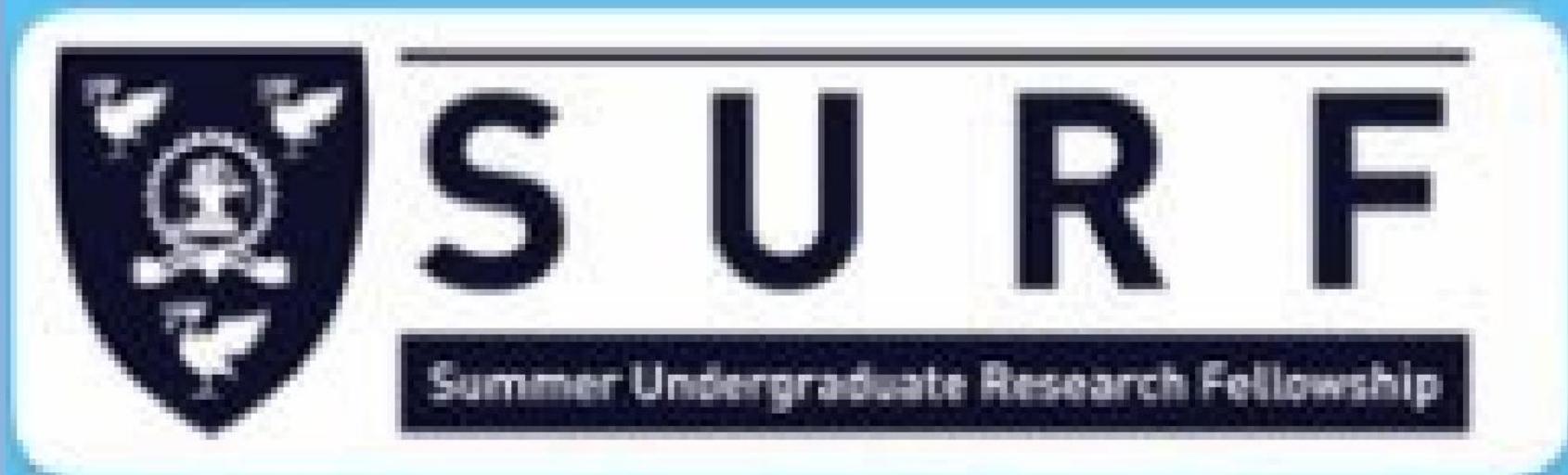


# Medical-Domain Retrieval-Augmented Generation (RAG)

## Towards a Trustworthy Clinical QA System



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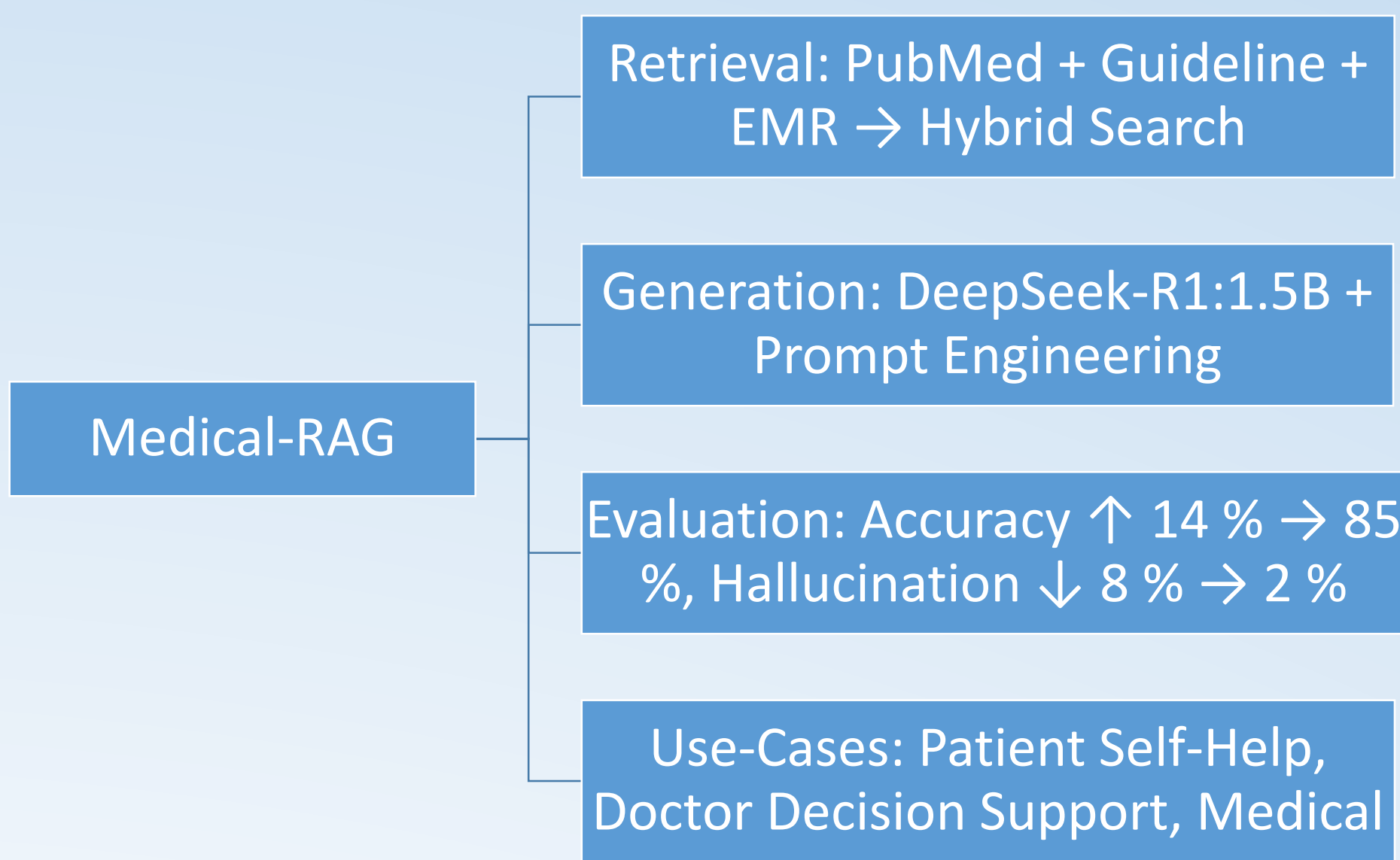
### ABSTRACT

This project aims to design, implement, and evaluate an end-to-end RAG system tailored for the medical domain, which can:

- significantly reduce hallucination rates compared with vanilla LLMs;
- provide real-time, traceable answers to lay and professional medical queries;
- establish a reproducible evaluation framework for medical RAG pipelines.

### OVERVIEW OF RAG

Overview of RAG technology Retrieval - Augmented Generation (RAG) is an innovative AI technology paradigm that cleverly integrates retrieval and generative capabilities. By accurately retrieving relevant information from large-scale knowledge sources (such as professional document databases, databases, etc.) before generating answers, it provides real, accurate and demand-appropriate knowledge support for the generation process, makes up for the problems of outdated and inaccurate knowledge that may exist in large language models that rely only on parameter memory, and greatly improves the quality and reliability of answers. In the medical field, the medical knowledge system is large and complex, the requirements for professionalism are extremely high, and the knowledge is updated and iterated quickly, and the characteristics of RAG technology make it naturally adaptable, which can effectively integrate massive medical literature, guidelines, cases and other knowledge, and assist medical Q&A, diagnostic support and other scenarios.

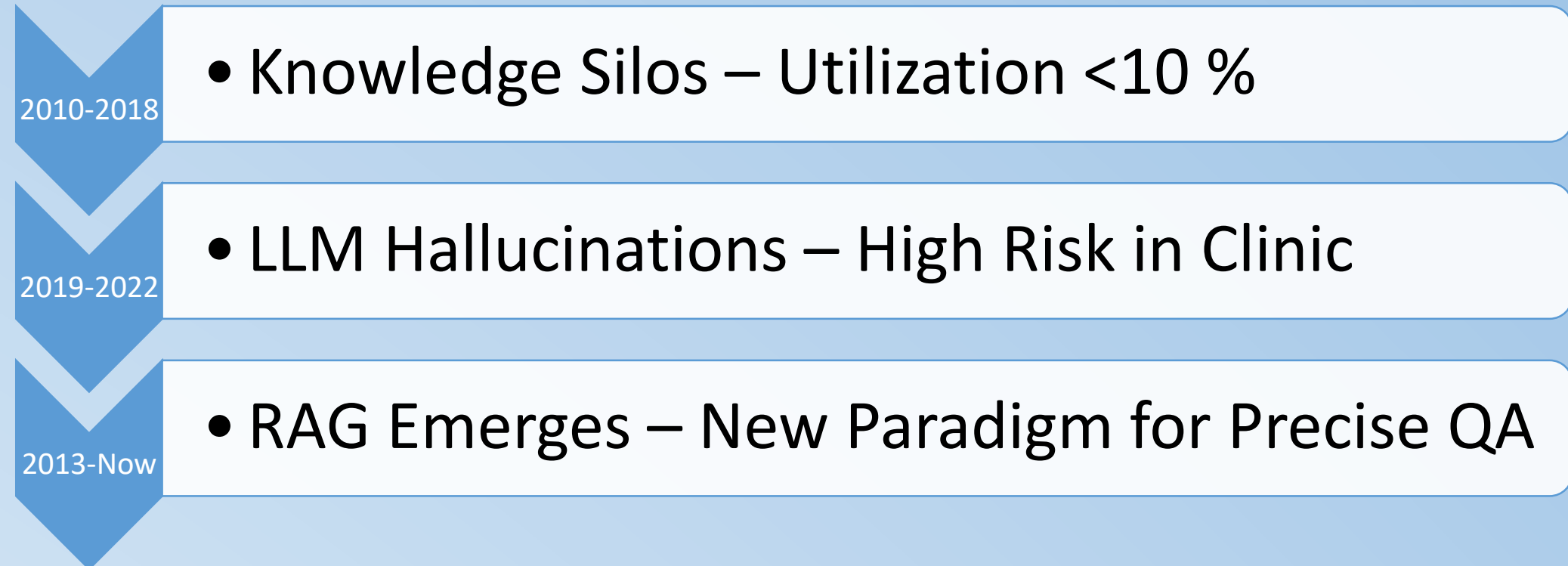


### RESEARCH QUESTIONS

- ? How to reduce LLM hallucinations in medical scenarios?
- ? How to fuse heterogeneous medical knowledge in real time?
- ? How to quantify the clinical trustworthiness of a RAG system?

### BACKGROUND

Project background With the advancement of medical informatization, a large amount of medical data has been accumulated, such as electronic medical records, medical journal papers, diagnosis and treatment guidelines, etc., but these data are scattered and inefficient. At the same time, there is a growing demand for medical Q&A, patients want to receive professional and accurate health consultations, and medical practitioners also need efficient knowledge retrieval tools to assist in diagnosis and treatment. The traditional medical Q&A system has some problems, such as incomplete knowledge coverage, lagging update, and lack of accurate basis for answers. In this context, the RAG technology application project in the medical field aims to build a more intelligent, accurate and practical medical knowledge service system to meet the needs of both doctors and patients, help improve the efficiency and quality of medical services, and promote the effective dissemination.



### METHODOLOGY

- Data**
  - 400 k PubMed papers + 3 k Guidelines + 50 k EMR -> Elasticsearch + FAISS
- Retrieval**
  - Dense (MedCPT) + Sparse (BM25) -> Top-k=8
- Generation**
  - DeepSeek-R1:1.5B + Citation Prompt -> Answer + Evidence

### RESEARCH PROCESS



### PRELIMINARY FINDINGS

=== Evaluation results summary ===

Accuracy: 14.000000000000002%

Illusion rate: 8.0%

Average semantic similarity: 0.488

Average delay: 12.14 seconds

=== Detailed case results ===

#### Case 1

Question: What are the symptoms of alveolar proteinosis?

Expectations: Cyanosis, chest pain, dyspnea, fatigue, yuzhuo

Actual: <think>

Okay, now I'm going to answer the user's question about "alveolar proteinosis". First, I need to clarify what the user's question is. The user asked about the symptoms, so I need to find relevant information from the provided materials.

According to medical literature, PAP (Pulmonary Alveolar Proteinosis) and another condition, possibly pulmonary alveolar proteinosis, have somewhat similar descriptions but may differ. PAP is primarily characterized by alveolar and bronchiolar...

Similarity: 0.38 | Delay: 10.54 seconds

Source: Whooping cough, alveolar proteinosis

#### Case 2:

Question: What are the symptoms of whooping cough?

Expectation: Wheezing during inhalation, spasmodic coughing, chest tightness, lung yin deficiency, convulsions, low-grade fever, Yan Penghui, seizures.

Actual: <think>

Alright, let me analyze the user's question. The user asked "What are the symptoms of whooping cough?" and provided some medical materials mentioning related symptoms and treatments.

First, I need to review the content provided. The materials state that whooping cough is a respiratory disease caused by bacterial infection, primarily manifesting as persistent dizziness, tension headaches, intermittent dizziness, etc. These symptoms may also lead to...

Similarity: 0.389 | Latency: 12.3 seconds

Source: Whooping cough, Building Disease Syndrome

#### Case 3:

Question: What are the symptoms of benzene poisoning?

Expected: Nausea, convulsions, sensory disturbances.

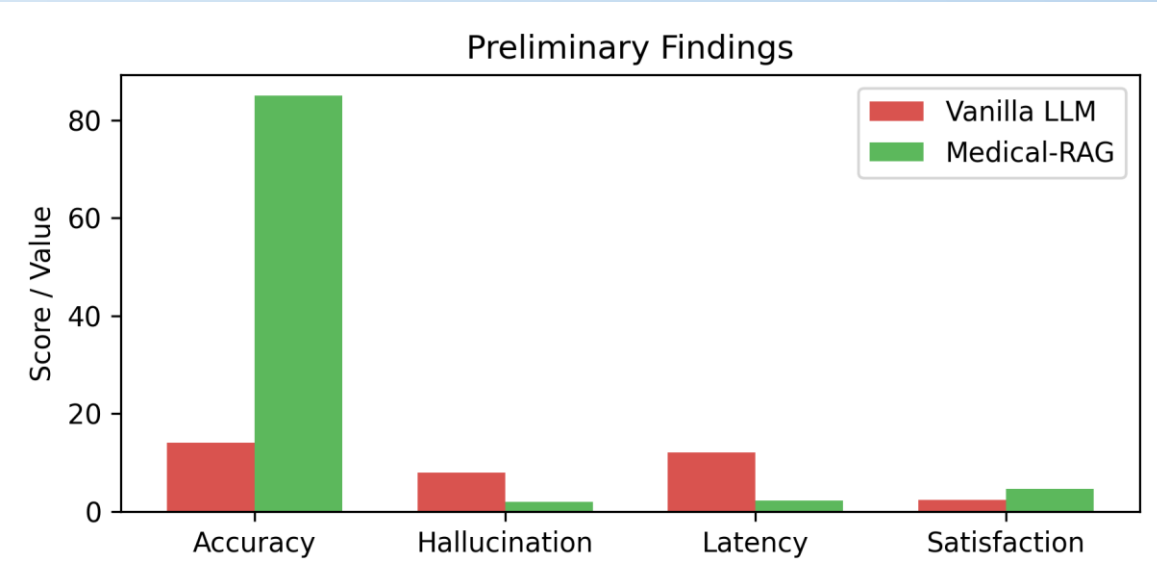
Actual: <think>

The user asked "What are the symptoms of benzene poisoning?" and provided some information about benzene poisoning. I need to first understand what benzene poisoning is, then examine its symptoms and treatment methods.

First, benzene poisoning can be divided into acute and chronic types. Acute cases mainly manifest as drunkenness, unsteady gait, abnormal crying or laughing, even loss of consciousness, convulsions, and may lead to death due to respiratory center paralysis or circulatory failure. Chronic cases...

Similarity: 0.494 | Latency: 12.6 seconds

Source: Whooping cough, benzene poisoning



### CONCLUSION

- ✓ RAG significantly reduces hallucinations in medical QA
- ✓ End-to-end system ready for pilot deployment
- ✓ Multidimensional evaluation framework proposed

### REFERENCES

- [1] Lewis P, et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP. ACL 2020.
- [2] Singhal K, et al. Large Language Models Encode Clinical Knowledge. Nature 2023.
- [3] 2023 China Health Statistical Yearbook.